

SOFIA PETRUSENKO

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PROFESSIONAL SUMMARY

Engineer building production LLM systems, with a biomedical research background in ageing biology. Ships end-to-end: FastAPI / PostgreSQL / Redis services, Next.js front ends, LLM orchestration with evaluation-first design — deployed and operating in production since 2025. Research training across the UK, Portugal, Germany and Ukraine (autophagy, proteostasis, longevity). Target: AI engineering in healthtech, longevity and scientific tooling, where measurement integrity matters as much as shipping speed.

SELECTED PROJECTS

blotquant — QC-first western blot densitometry | 2026 – v1.0 in progress

github.com/sofiapetrusenko/blotquant · case study: sofiapetrusenko.dev/work/blotquant

- Python image-analysis tool built on the principle that a bad measurement is a refusal, not a warning: three-state QC model (pass / flagged / blocked); blocked lanes produce no numeric output and invalidate any ratio computed from them.
- FastAPI service (analyze / results endpoints), OpenCV + scikit-image pipeline, pytest suite and CI; independent validation against Fiji pre-registered before any data was seen.

lifespan-extract — structured longevity data from biomedical literature | 2026 – in development

github.com/sofiapetrusenko/lifespan-extract · case study: sofiapetrusenko.dev/work/lifespan-extract

- LLM extraction pipeline measured against a hand-labeled gold set built before the pipeline existed; per-field evaluation metrics and quote verification of every extracted value against source text.
- Engineered as agentic loops (implementer → reviewer subagents) with human-gated eval design and gold-set labeling — the model is never allowed to grade its own homework.

TECHNOLOGY & COMMERCIAL EXPERIENCE

Founder & Engineer — AI Content Automation Platform | Independent venture | 2025 – present (operating)

- Designed, built and operate a full-stack platform for automated media production and distribution: LLM-driven generation, text-to-speech, video assembly, publishing via the YouTube Data API, analytics ingestion and monitoring.
- Architected a seven-stage processing pipeline (script → voiceover → asset sourcing → assembly → subtitling → metadata → thumbnail) orchestrated through asynchronous background workers.
- Stack: Python / FastAPI, PostgreSQL, Redis with RQ workers, Next.js front end, Docker, deployed on DigitalOcean; integrated third-party APIs for LLM inference, speech synthesis and stock media.
- Runs a multi-channel network with human-in-the-loop editorial control, including a longevity & healthspan channel drawing directly on biomedical subject-matter expertise; extended to multi-language output (Spanish) from the same pipeline.
- Sanitized platform code (architecture, pipeline, tests) open-sourced at github.com/sofiapetrusenko/media-automation-platform.

AI Engineer — Conversational Agent with Persistent Long-Term Memory | Independent project | 09/2025 – 12/2025

- Built a reflective journaling-style conversational agent on an Obsidian Markdown knowledge base, with durable memory across sessions rather than single-conversation context.
- Developed the memory layer end-to-end: structured note schema, automated extraction of salient information, and retrieval of relevant prior context at inference time.

Paid Acquisition & Performance Marketing | Independent / contract | 09/2025 – 2026

- Planned and optimised Google Ads and Meta campaigns against defined CPA and return targets; owned tracking, attribution and structured creative experiments interpreted statistically.

RESEARCH EXPERIENCE

Research Intern — Dr. Adrien Rousseau Lab, MRC PPU, School of Life Sciences | Dundee, UK | 09/2024 – 01/2025

- Investigated stress-induced protein phosphorylation in *S. cerevisiae* and its relationship to longevity; analysed experimental datasets in R and Python to identify patterns linked to yeast lifespan.
- Collaborated on experimental design and interpretation; contributed to drafting a manuscript on implications for ageing mechanisms.

Research Intern — Sandra Tenreiro & Paula Macedo Labs, NOVA Medical School | Lisbon, Portugal | 06/2023 – 08/2023

- Studied molecular mechanisms of ageing relevant to neurodegeneration and retinal disease with a focus on proteostasis, in models from yeast to human cells, brain and retinal organoids.
- Supported therapeutic research for Parkinson's disease, diabetic retinopathy and AMD in collaboration with clinicians.

Research Technician — Dr. James Murray Lab, Institute of Life Sciences | UK | 05/2022 – 09/2022

- Investigated autophagy signalling and its alteration during ageing and in disease contexts including cancer, neurodegeneration and diabetes; contributed to work on autophagy in immunity and SARS-CoV-2-related infection.

Research Intern — Prof. Dr. Iain D. Couzin, Max Planck Institute of Animal Behaviour | Konstanz, Germany | 03/2022 – 05/2022

- Researched collective animal behaviour and social influence across biological systems, combining experimental and theoretical approaches.

Laboratory Assistant — Anatoliy I. Dragan Lab | Kyiv, Ukraine | 02/2021 – 09/2021

- Supported an experimental study of DNA hydration using Pressure Perturbation Scanning Microcalorimetry.

CORE SKILLS

Software Engineering & AI: Python, FastAPI, TypeScript / Next.js, PostgreSQL, Redis / RQ, REST APIs, LLM orchestration and prompt engineering, evaluation design, RAG and long-term memory, agentic workflows (Claude Code), Docker, Git, CI, Vercel / DigitalOcean deployment

Data & Bioinformatics: R, Python (pandas, NumPy), statistical analysis and visualisation, OpenCV / scikit-image, BLAST, UniProt, ClustalX, Jalview

Molecular & Cell Biology: qPCR, western blotting, ELISA, immunoprecipitation, mammalian and bacterial cell culture, DNA/protein extraction and cloning workflows; confocal, light and electron microscopy

Growth & Performance Marketing: Google Ads and Meta campaign architecture, conversion tracking and attribution, funnel and cohort analytics

Languages: English (fluent), Ukrainian (native), Russian (native), Portuguese (learning)

EDUCATION

BSc (Hons) Biomedical Sciences — University of Dundee | Scotland, UK | 09/2022 – 05/2025

- Placement and project work focused on ageing biology, autophagy, proteostasis and longevity mechanisms.

BSc Biology — Taras Shevchenko National University of Kyiv, Institute of High Technologies | Kyiv, Ukraine | 09/2020 – 04/2022

ADDITIONAL

- Training: ICH E6 Good Clinical Practice (GCP) Investigator Site Training.
- Comfortable working autonomously and shipping end-to-end — from specification through build to live operation.